

Efforts and progress in detection methodologies of *Giardia* and *Cryptosporidium* from water

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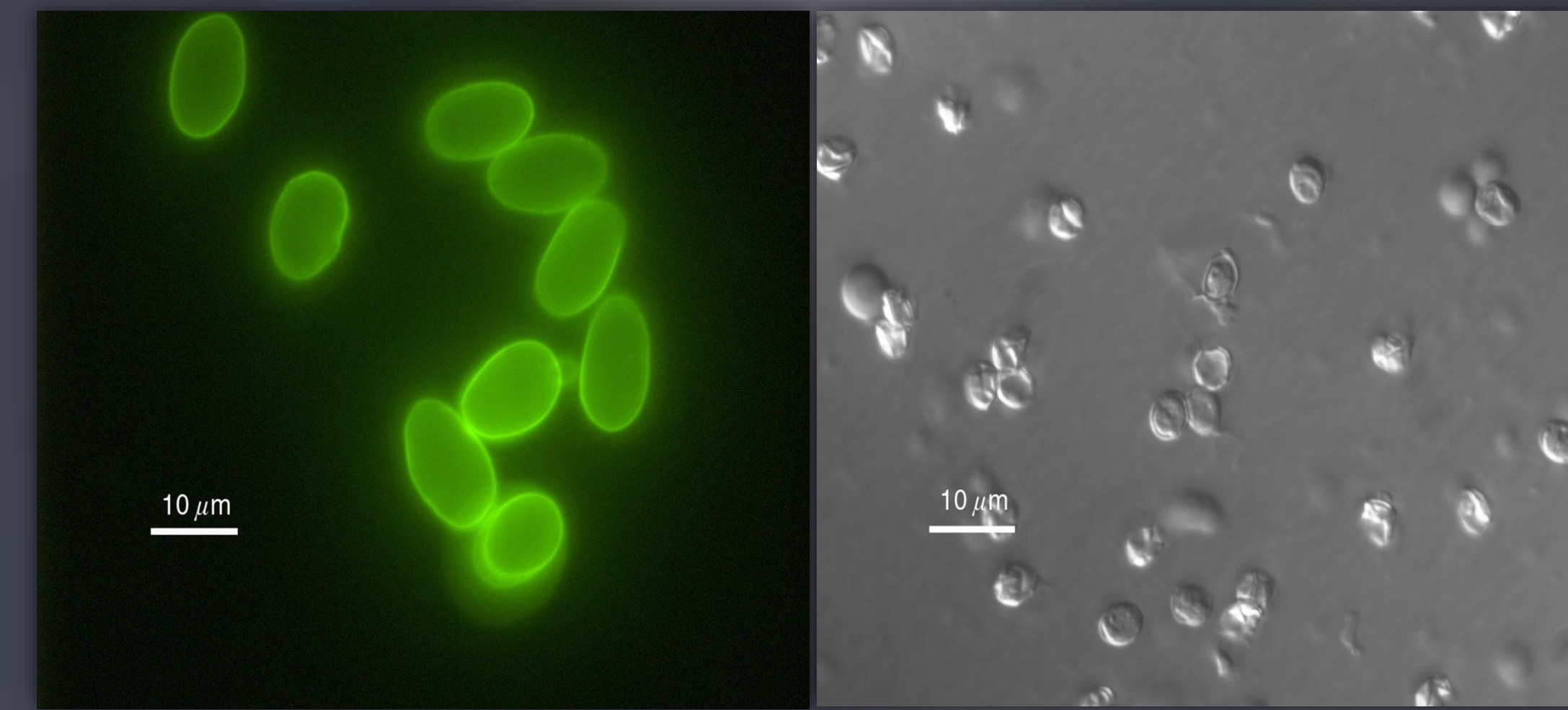
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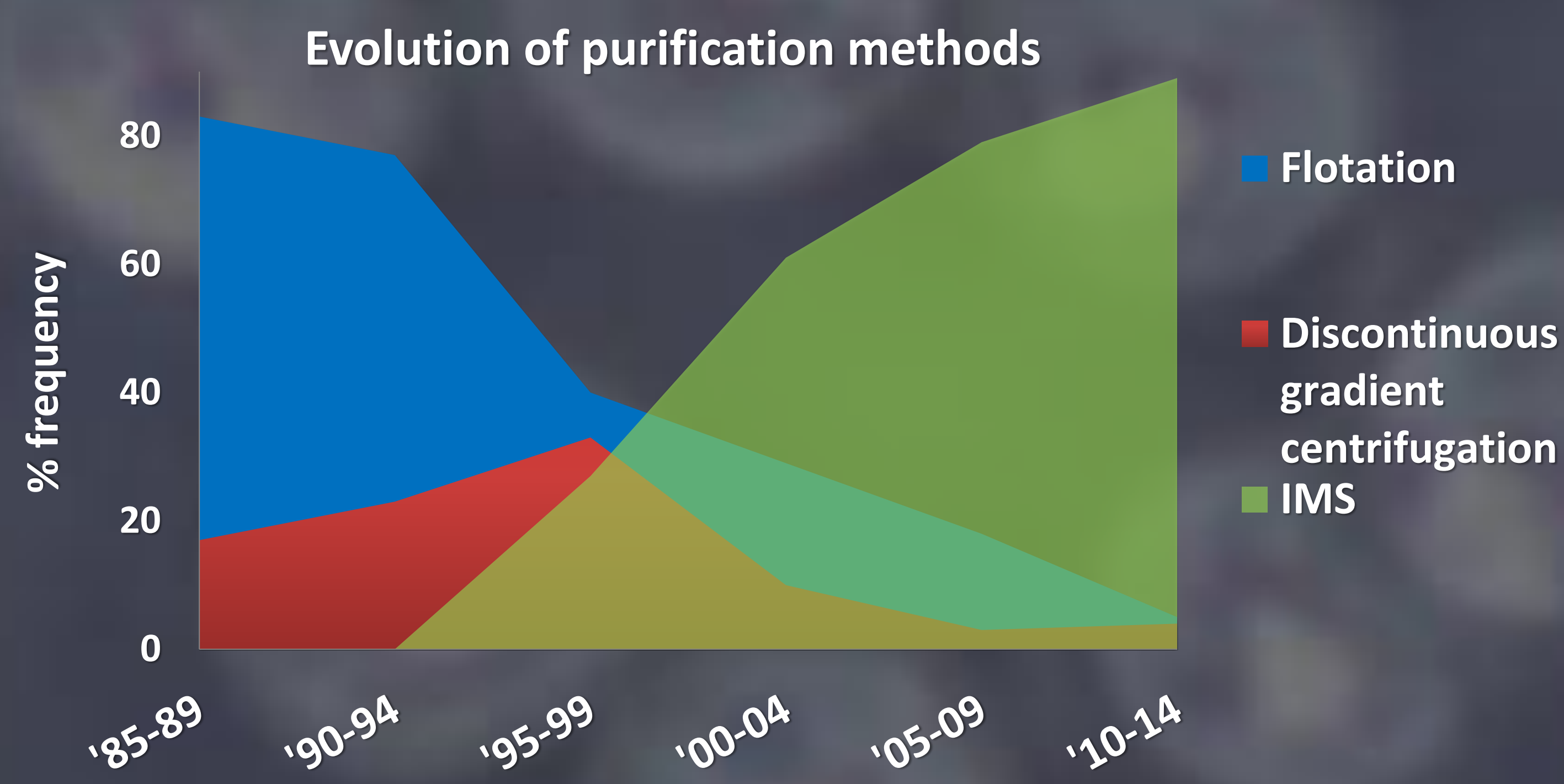
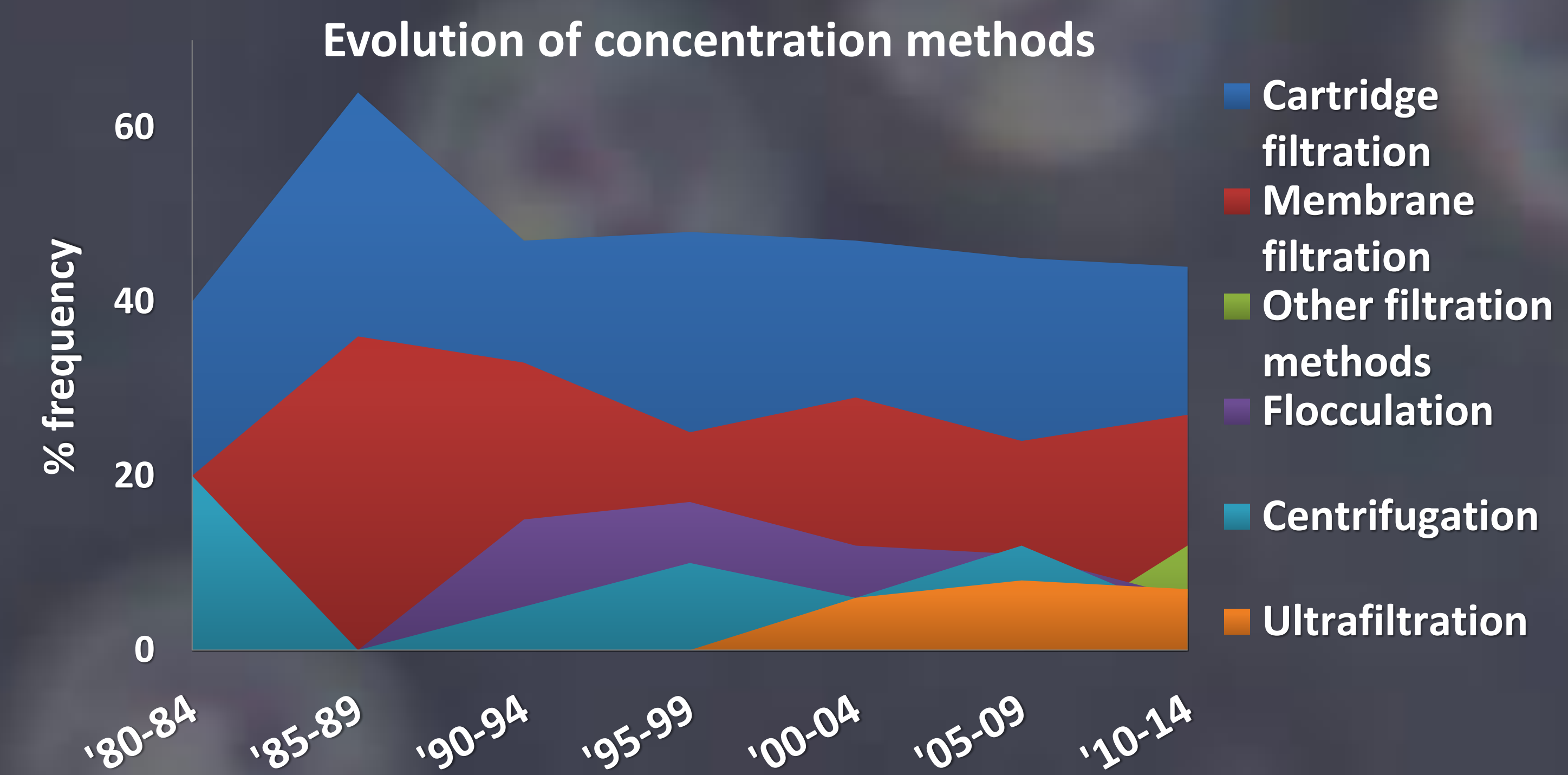
Plastiras lake, Greece

The parasitic protozoa *Cryptosporidium* and *Giardia* constitute the leading causes of waterborne enteric disease outbreaks worldwide. Detection methods from water have changed considerably since the late 70's. From the early years of *Cryptosporidium* and *Giardia* detection in water research has been published from 44 countries. The methodology typically includes the following steps:



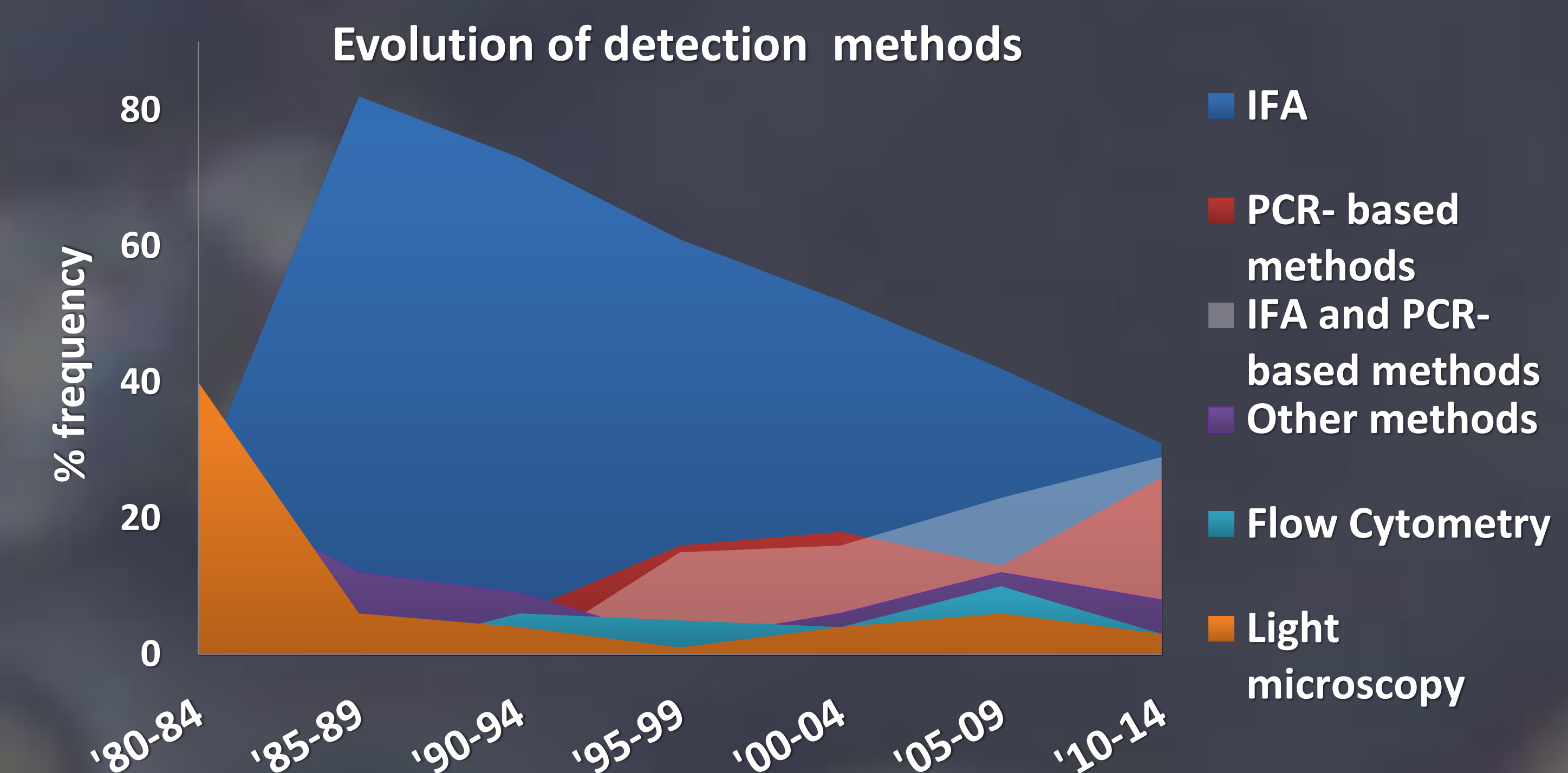
G. lamblia cysts by IFA (left) and *C. parvum* oocysts by DICM (right)
(Photo Credit: H.D.A Lindquist, U.S. EPA)

CONCENTRATION: Among the first concentration methods were membrane and cartridge filtration, which quickly became the most prominent ones. Cartridge, unlike membrane filters, enabled researchers to sample large volumes regardless of the turbidity levels. Vortex-flow and cross-flow filtration, calcium carbonate, aluminium and ferric sulphate flocculation, and continuous flow centrifugation (CFC) constituted the alternatives. Despite the advent of ultrafiltration techniques, cartridge filtration (approved by USEPA Method 1623 in 1999) remained the most popular choice, followed by membrane filtration. New promising techniques (ARAD microfiber filtration, micro/nano-filtration) were introduced more recently. Overall, in the last 35 years, filtration makes up for 83% of the concentration methods (cartridge 45%, membrane 27%, ultrafiltration 6%, other techniques 5%), followed by flocculation (10%), and centrifugation (7%).

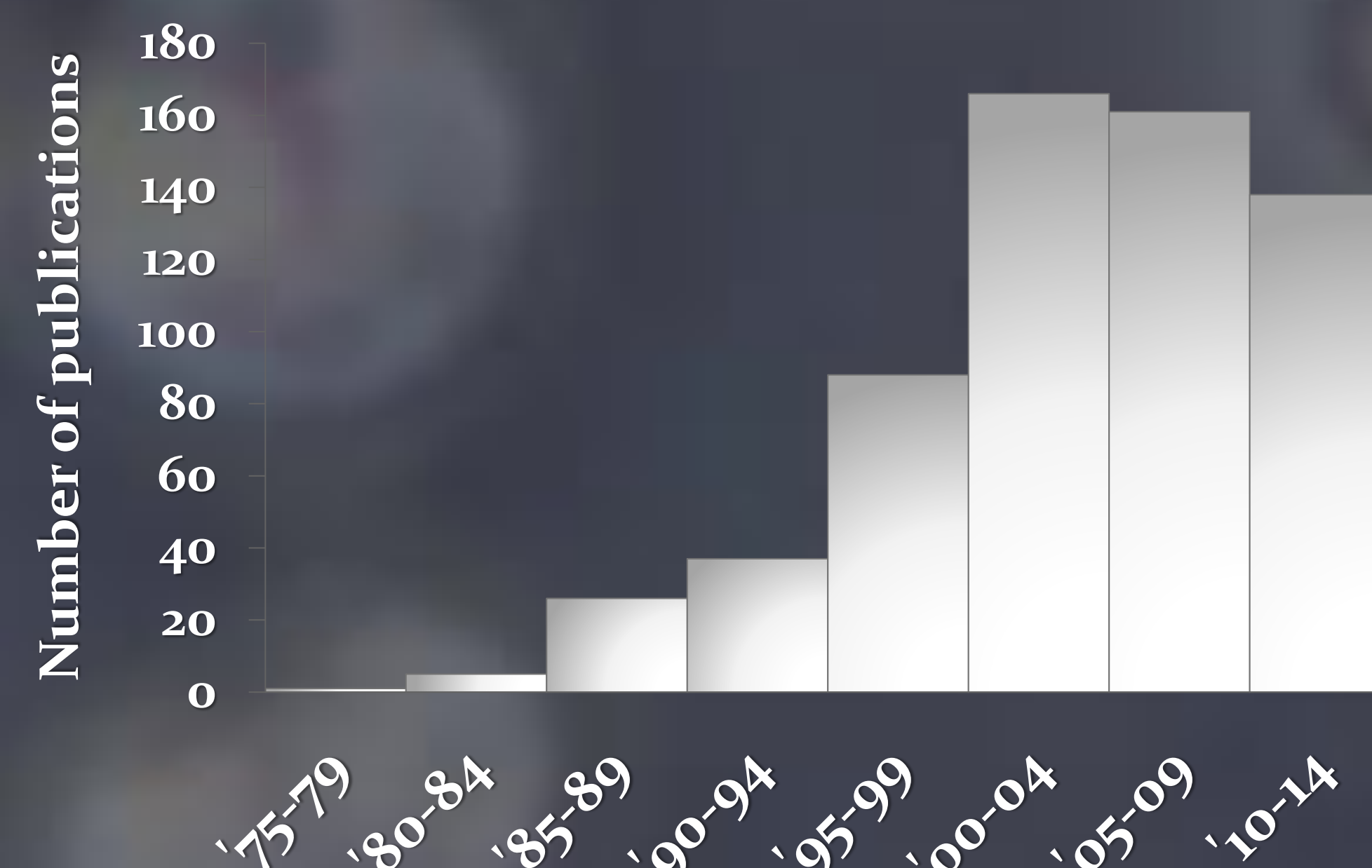
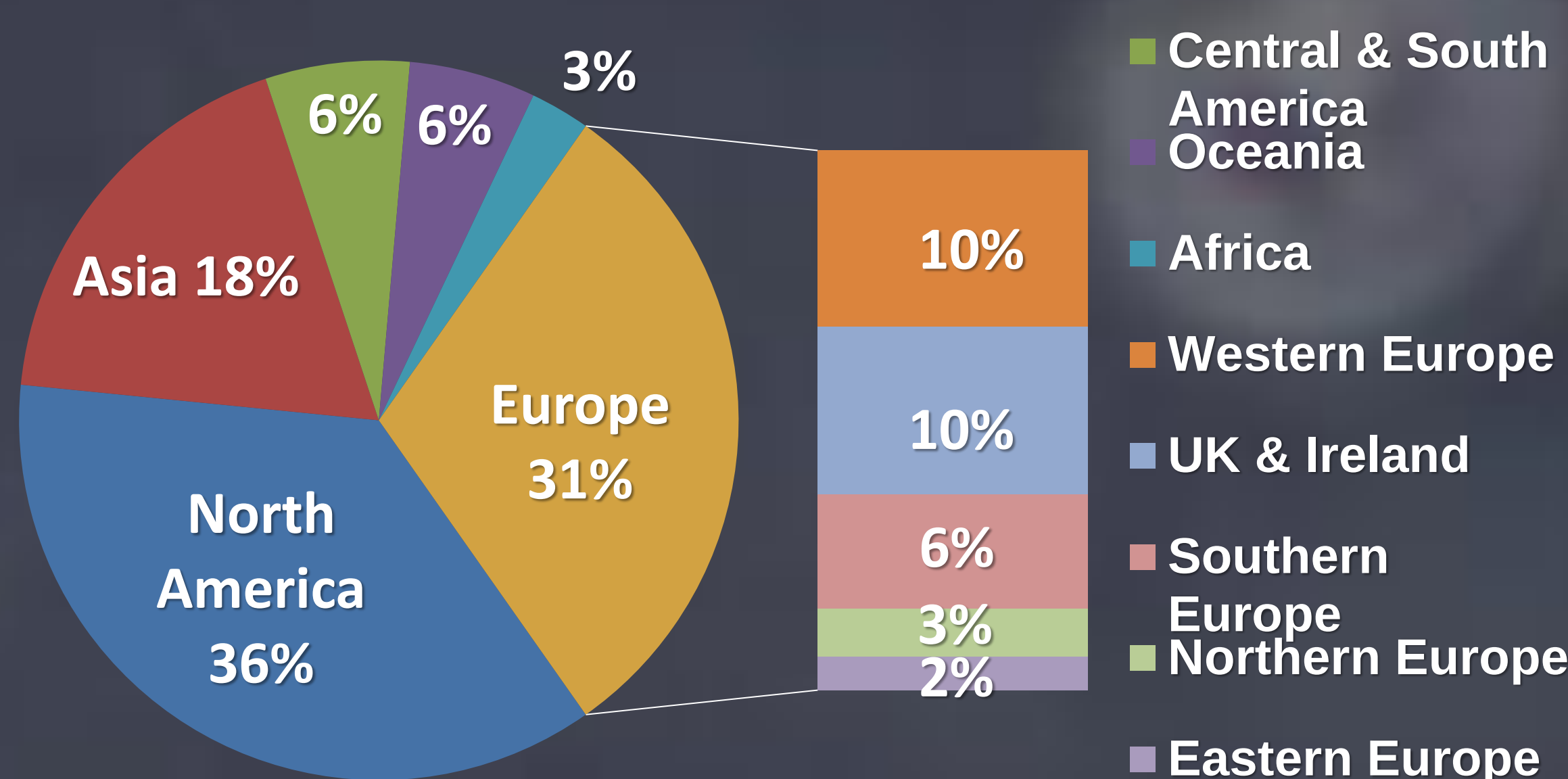


PURIFICATION: In 1985 density gradient separation methods were the first to be employed. Sucrose, percoll/sucrose, zinc sulfate and potassium citrate flotation, as well as percoll/sucrose discontinuous gradient centrifugation were employed for the next 12 years. Magnetic-based separation was introduced in 1993 and immunomagnetic capture in 1997, which would come to be known as immunomagnetic separation (IMS). IMS has since steadily become the most prominent purification method for both parasites, and was soon integrated into Method 1623. The use of density gradient separation techniques has inevitably declined ever since. Overall, IMS holds the lion's share of the purification methods used, with 67%. Flotation (21%) and discontinuous gradient centrifugation (12%) are still applied to this day for the purification of (oo)cysts from water samples.

DETECTION: The detection of (oo)cysts during the early years was exclusively performed by brightfield, fluorescence and phase contrast microscopy. The following years DIC microscopy (DICM) and staining techniques were used to assist with sample observation, while immunofluorescence assay (IFA) continued to be popular. In the early 90's IFA was first combined with DAPI staining, and Flow Cytometry was first utilized. Later, a variety of PCR methods emerged and were combined with sequencing to allow the differentiation of species. After 1999 many researchers employed the proposed USEPA procedure of detection by IFA, DAPI staining and DICM. In the last 10 years, despite the advent of new detection methods, such as LAMP, biosensors, and novel spectroscopic and microscopic techniques, this pattern hasn't undergone drastic changes. Overall, since the late 70's IFA has been the most prominent detection method (42%), with PCR-based methods (20%) and the combination of them (18%) being the most common techniques for the detection of *Cryptosporidium* and *Giardia* from water.



PUBLICATIONS: The distribution of the publications into 5-year time periods and the regional distribution during 1975-2014 are presented. The data were collected from 622 publications.



USEPA Method 1623 was revised in 2001, 2003, 2005, and updated (1623.1) in 2012, its core procedure has remained unchanged, and involves concentration by cartridge filtration, purification by IMS, and detection by IFM, DAPI and DICM. In 2005 portable CFC was accepted as an alternative to cartridge filtration.

COSTS: The overall cost for the process of a single water sample with Method 1623 is above € 250, and can rise to € 500 or even more including positive controls for the calculation of the recovery efficiency. Since 1999 method 1623 has been used to isolate *Cryptosporidium*, *Giardia* or both in approximately one third (32%) of the papers published. Except for the considerable costs, another drawback of method 1623 is the recovery efficiencies.

RECOVERY EFFICIENCY: Approximately 40% of the papers that have used Method 1623 have calculated the recovery efficiency. Reported efficiencies range depending on the volume of the sample, turbidity, as well as seeding levels, and mean recoveries vary greatly, ranging from below 10% for high turbidity samples to above 75% for tap water for both parasites. According to Method 1623.1, matrix spike recoveries are below 30% almost 28% of the time for *Cryptosporidium* and 17% for *Giardia*. It has also been criticized as time consuming, due to the long analysis time per sample.

CONCLUSIONS: A satisfactory method for the detection of both parasites from water would have to be efficient, cheap and fast. The methods that have evolved over the last 35 years lack in all three features. Efficiency leaves much to be desired, and the cost of equipment and lab consumables is prohibitive for routine laboratories of all but the most affluent countries. Time required for detection, given the urgency of management interventions when it concerns drinking water, is long. Based on the great improvements in technique that have been reached, research is required on novel concepts, in order to develop a method that can be cheap, fast and more effective. International organizations, governmental and non-governmental, in liaison with ISO (International Organization for Standardization) and IEC (International Electrotechnical Commission) should collaborate closely to this end and standardize such a method, based on expertise in parasitological analysis. It is an urgent need to look for alternatives and new concepts for the development of a new, simple and effective methodology.